20-07-2026\

ACTIVITY

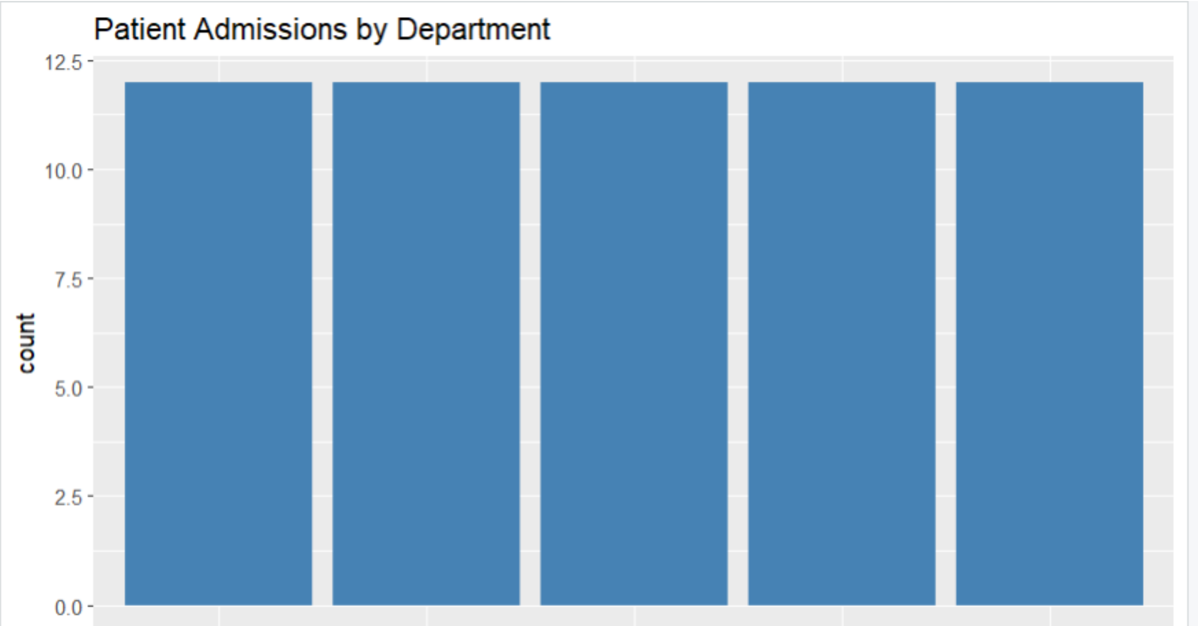
1.

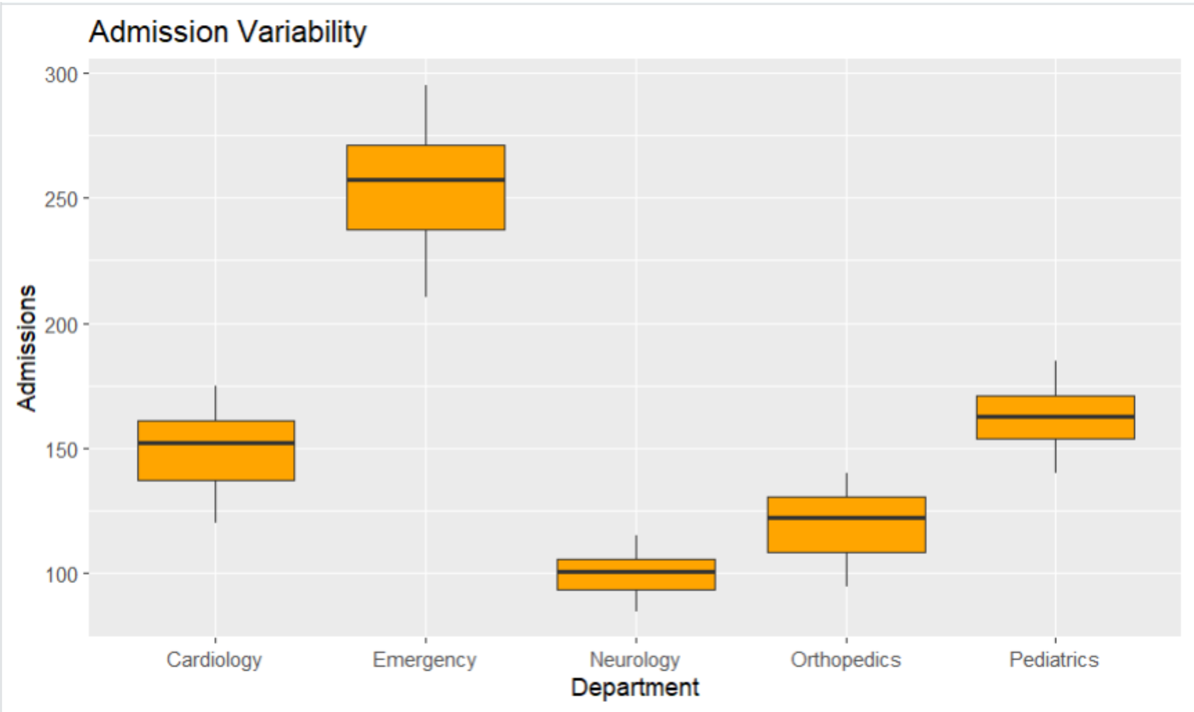
```
library(ggplot2)

library(dplyr)

library(plotly)

library(treemap)

# Read data

df <- read.csv("hospital_admissions.csv")

# Bar Chart

ggplot(df, aes(Department)) +

 geom_bar(fill="steelblue") +

 ggtitle("Patient Admissions by Department")

# Box Plot

ggplot(df, aes(Department, Admissions)) +

 geom_boxplot(fill="orange") +

 ggtitle("Admission Variability")

# Histogram

ggplot(df, aes(Admissions)) +

 geom_histogram(binwidth=5, fill="green", color="black") +

 ggtitle("Distribution of Admissions")

# Stacked Bar Chart

monthly <- df %>%

 group_by(Month, Department) %>%

 summarise(Admissions = sum(Admissions), .groups = "drop")

ggplot(monthly,

    aes(Month, Admissions, fill=Department)) +

 geom_bar(stat="identity") +

 ggtitle("Monthly Department Admissions")

# Treemap

treemap(df,
```

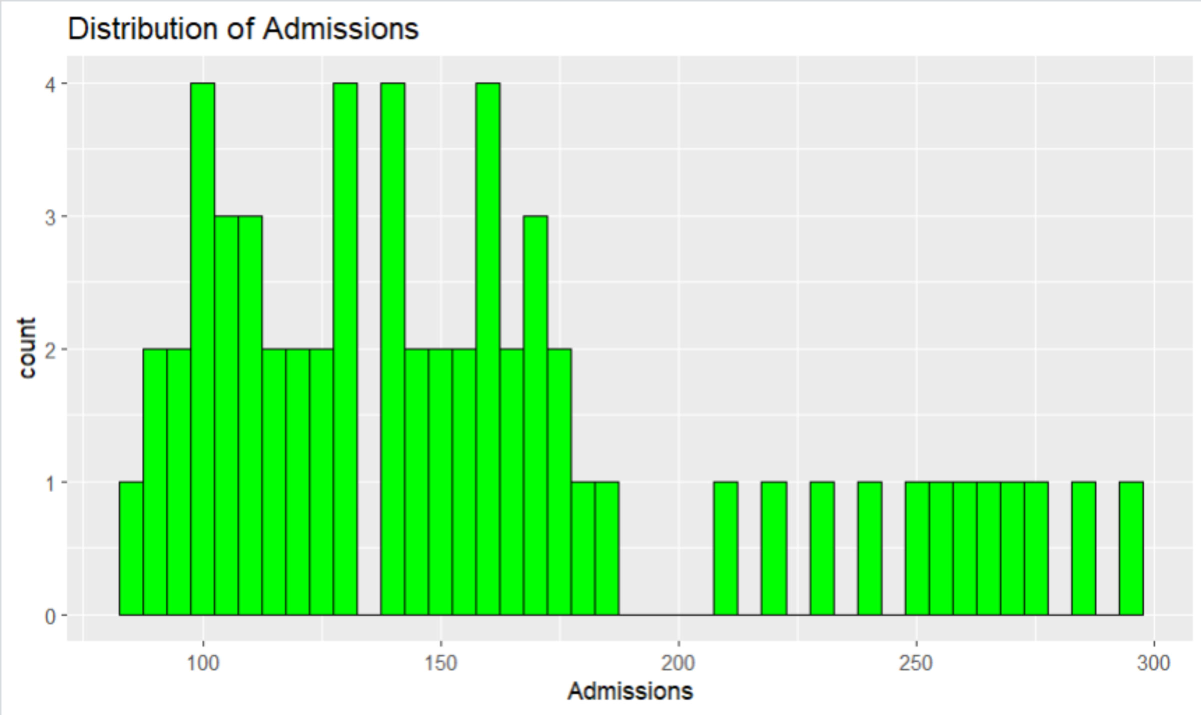
index="Department",

vSize="Admissions",

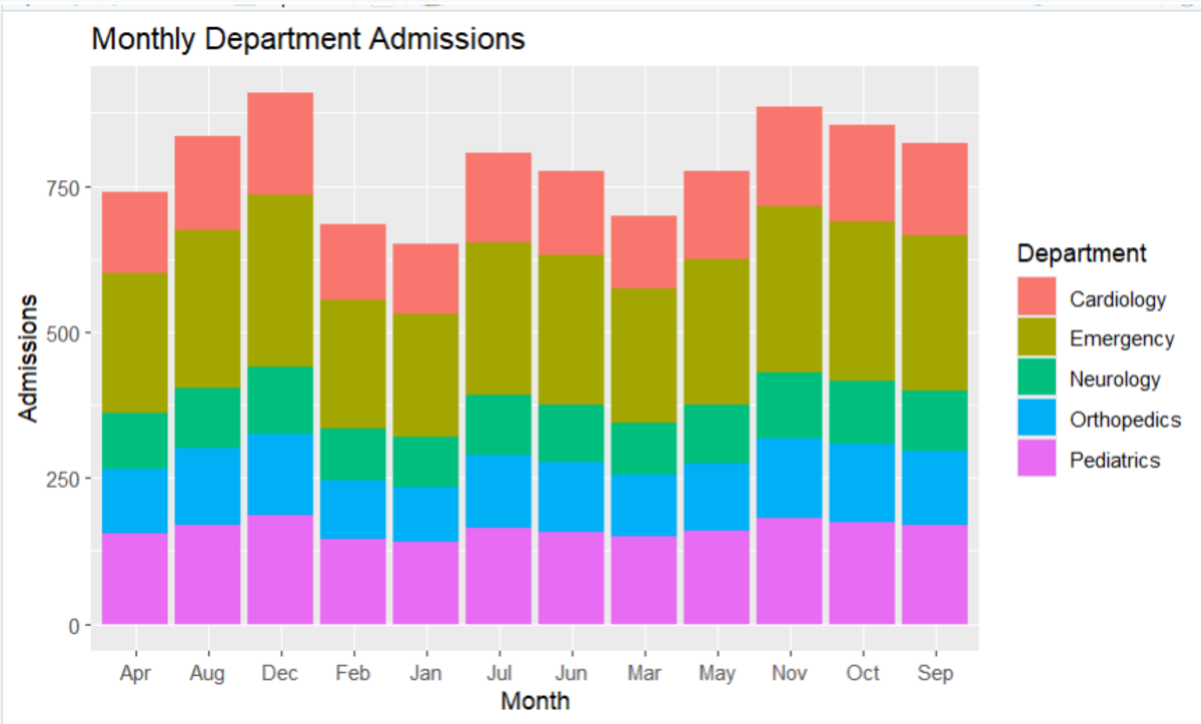
title="Department Workload")



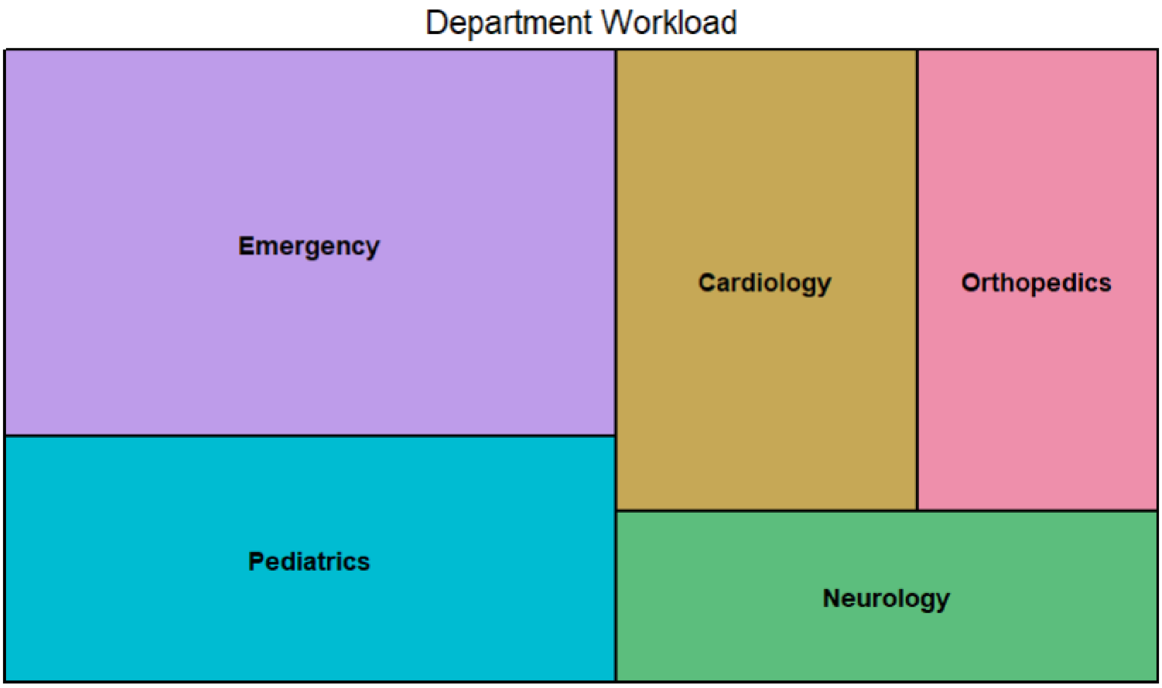




Distribution of Admissions
count
Admissions
0
1
2
3
4
100
150
200
250
300



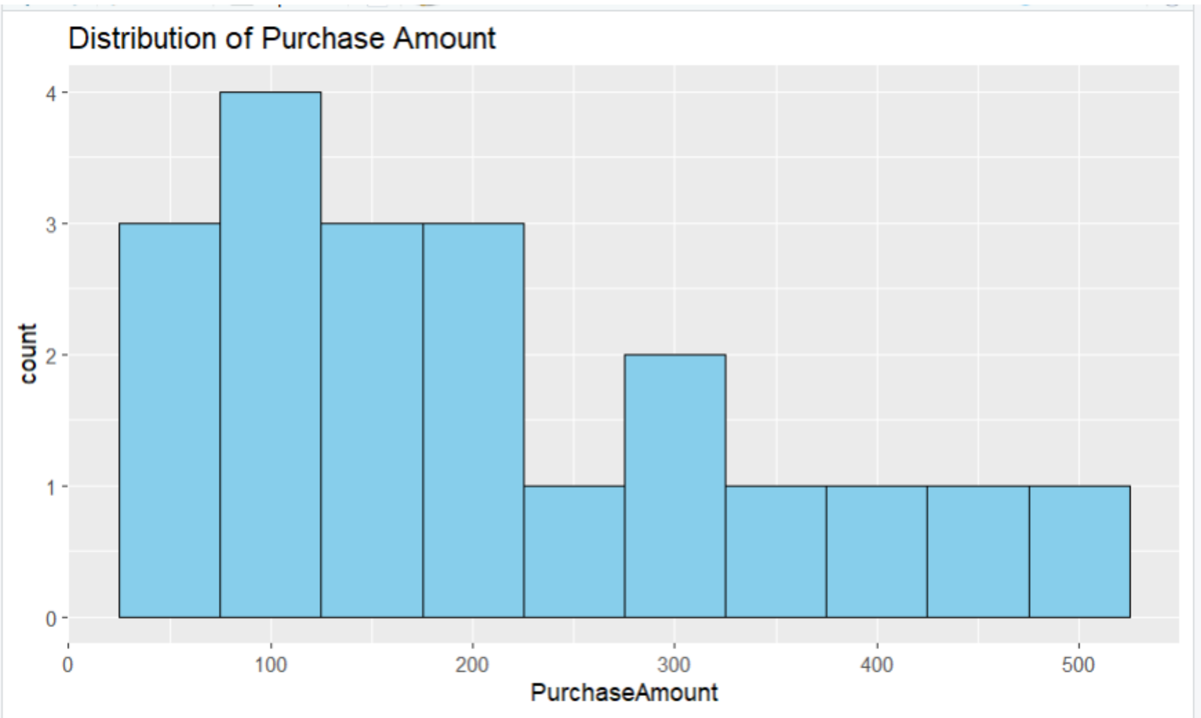
Monthly Department Admissions
Admissions
Month
0
250
500
750
Apr
Aug
Dec
Feb
Jan
Jul
Jun
Mar
May
Nov
Oct
Sep
Department
Cardiology
Emergency
Neurology
Orthopedics
Pediatrics

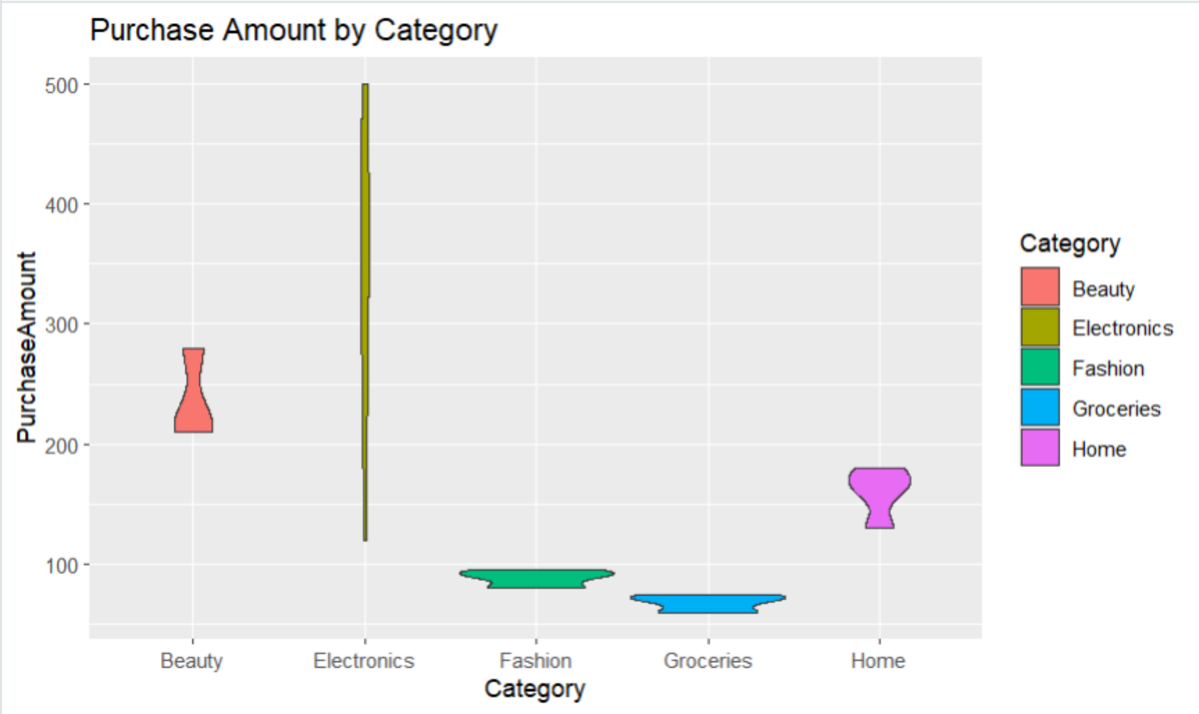


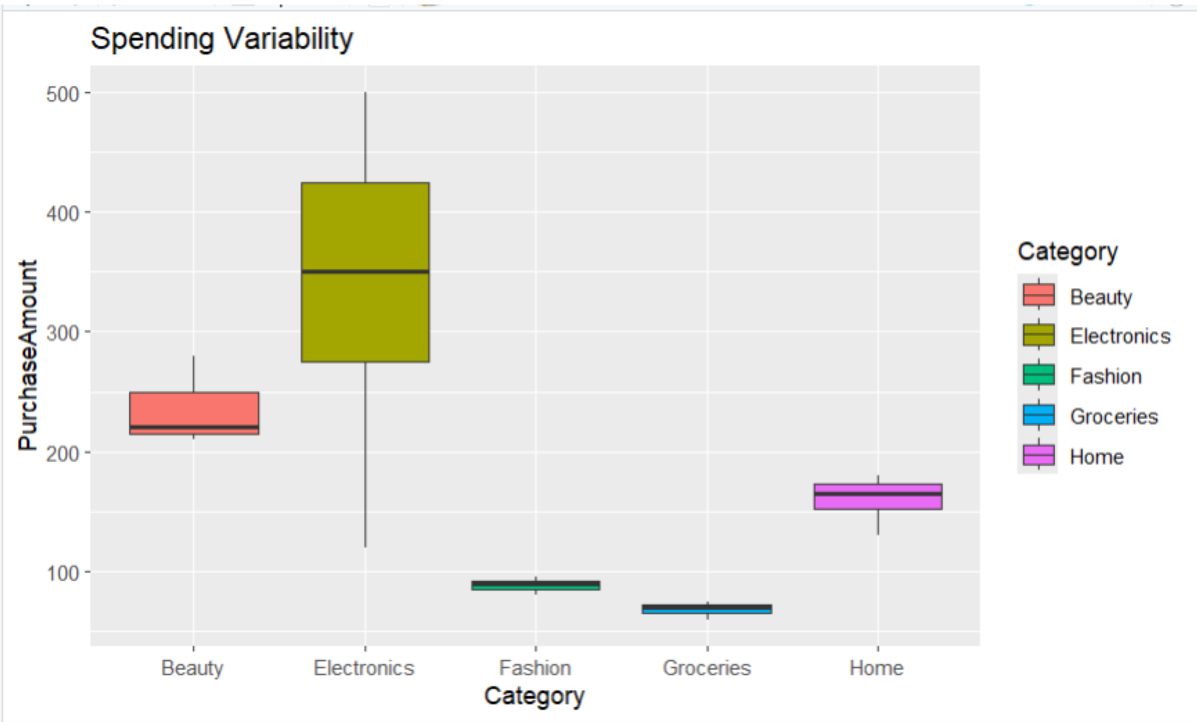
2.

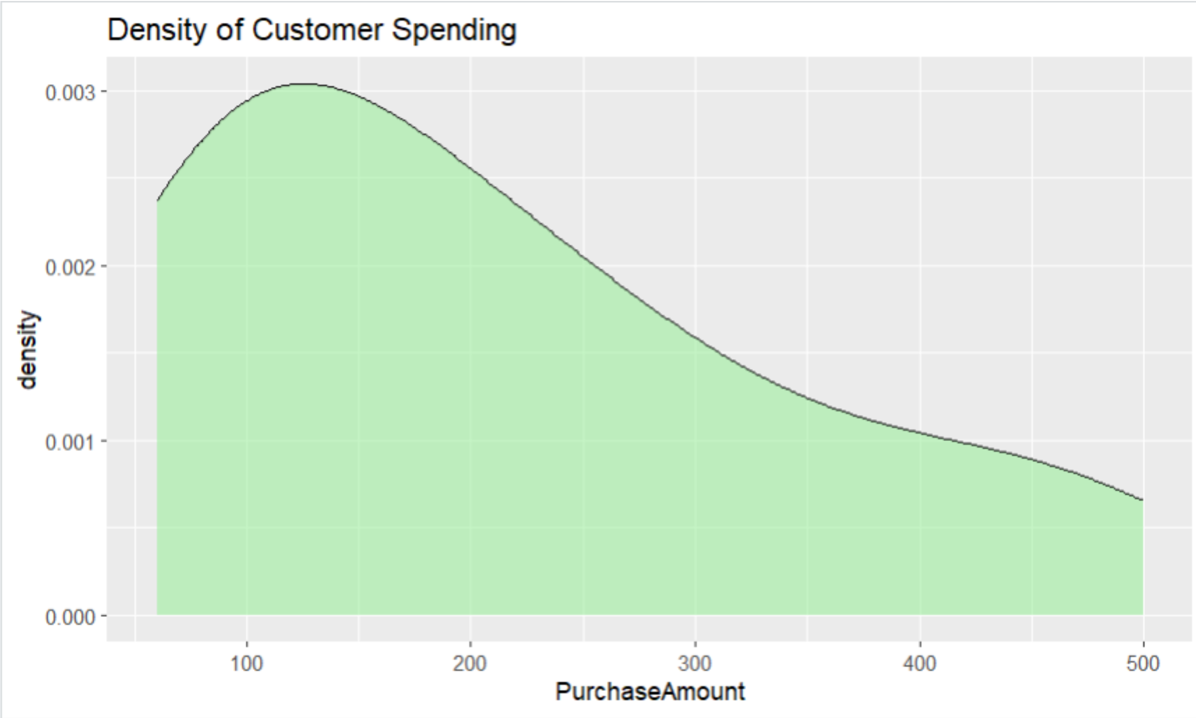
```
library(ggplot2)
library(dplyr)

# Read dataset
df <- read.csv("ecommerce_spending.csv")

# Histogram
ggplot(df, aes(PurchaseAmount)) +
  geom_histogram(binwidth=50, fill="skyblue", color="black") +
  ggtitle("Distribution of Purchase Amount")

# Violin Plot
ggplot(df, aes(Category, PurchaseAmount, fill=Category)) +
  geom_violin() +
  ggtitle("Purchase Amount by Category")

# Box Plot
```

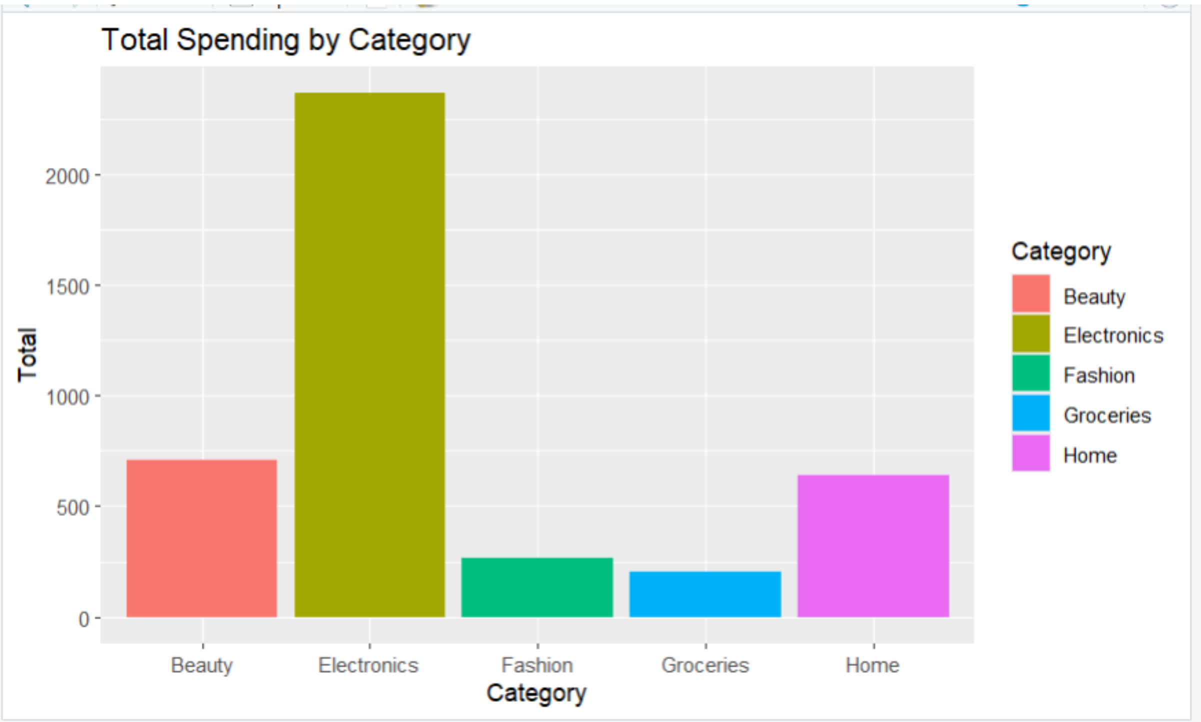
```r
ggplot(df, aes(Category, PurchaseAmount, fill=Category)) +
  geom_boxplot() +
  ggtitle("Spending Variability")

# Density Plot
ggplot(df, aes(PurchaseAmount)) +
  geom_density(fill="lightgreen", alpha=0.5) +
  ggtitle("Density of Customer Spending")

# Bar Chart
category_sales <- df %>%
  group_by(Category) %>%
  summarise(Total = sum(PurchaseAmount))

ggplot(category_sales,
     aes(Category, Total, fill=Category)) +
  geom_bar(stat="identity") +
  ggtitle("Total Spending by Category")
```




Purchase Amount by Category
PurchaseAmount
500
400
300
200
100
Beauty
Electronics
Fashion
Groceries
Home
Category
Category
Beauty
Electronics
Fashion
Groceries
Home


Spending Variability
PurchaseAmount
500
400
300
200
100
Beauty
Electronics
Fashion
Groceries
Home
Category
Category
Beauty
Electronics
Fashion
Groceries
Home



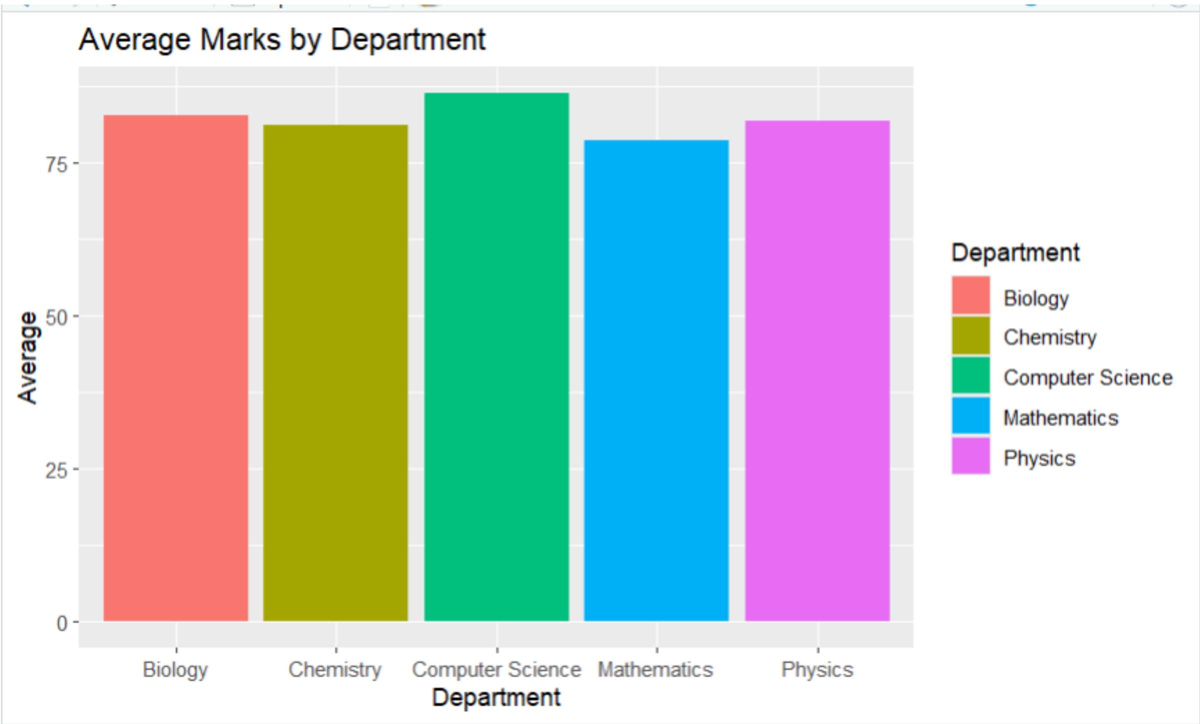


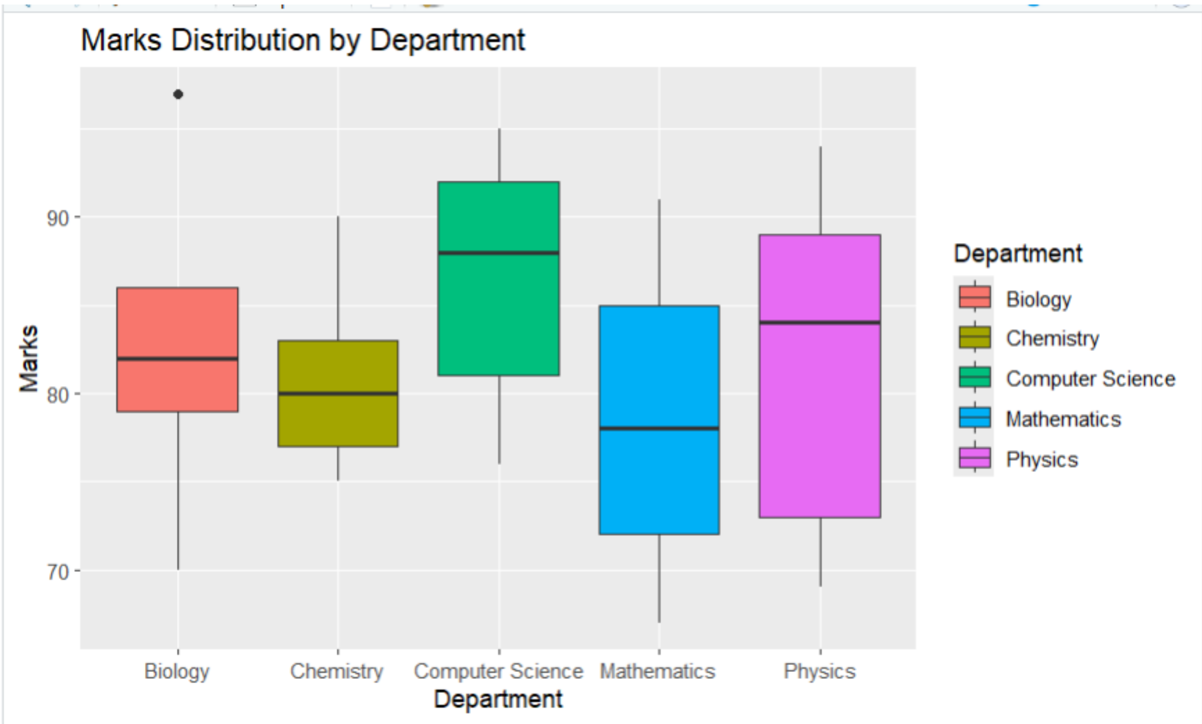
3.

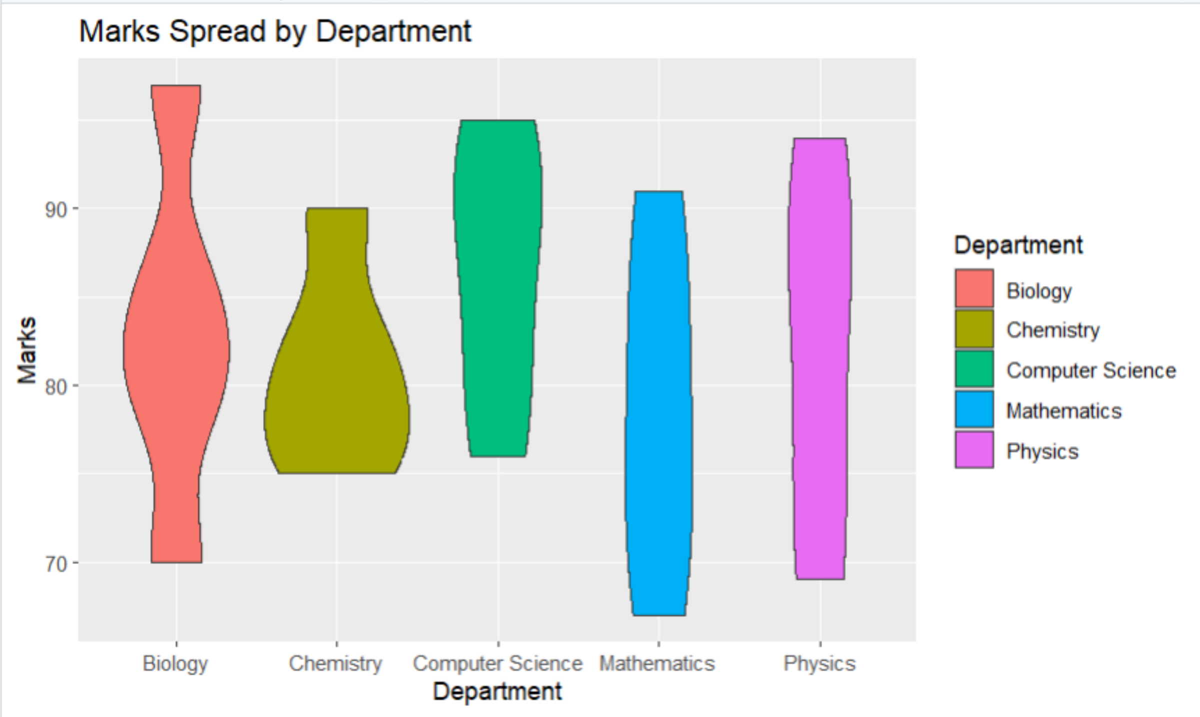
```
library(ggplot2)

library(dplyr)

library(plotly)

# Read dataset

df <- read.csv("university_marks.csv")
```

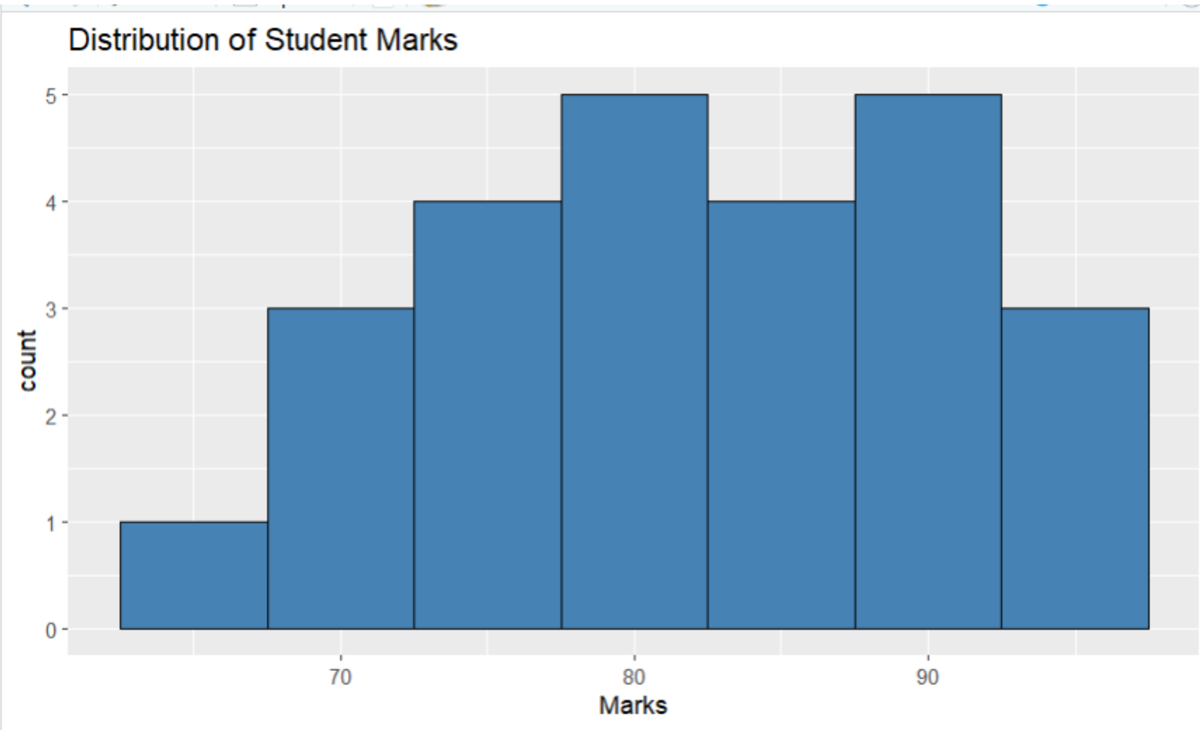
```r
# 1. Bar Chart
dept_avg <- df %>%
  group_by(Department) %>%
  summarise(Average = mean(Marks))
ggplot(dept_avg,
       aes(Department, Average, fill=Department)) +
  geom_bar(stat="identity") +
  ggtitle("Average Marks by Department")
# 2. Box Plot
ggplot(df,
       aes(Department, Marks, fill=Department)) +
  geom_boxplot() +
  ggtitle("Marks Distribution by Department")
# 3. Histogram
ggplot(df,
       aes(Marks)) +
  geom_histogram(binwidth=5,
                 fill="steelblue",
                 color="black") +
  ggtitle("Distribution of Student Marks")
# 4. Violin Plot
ggplot(df,
       aes(Department, Marks, fill=Department)) +
  geom_violin() +
  ggtitle("Marks Spread by Department")
# 5. Interactive Scatter Plot
fig <- plot_ly(df,
               x=~Marks,
               y=~Department,
               color=~Class,
               type="scatter",
```

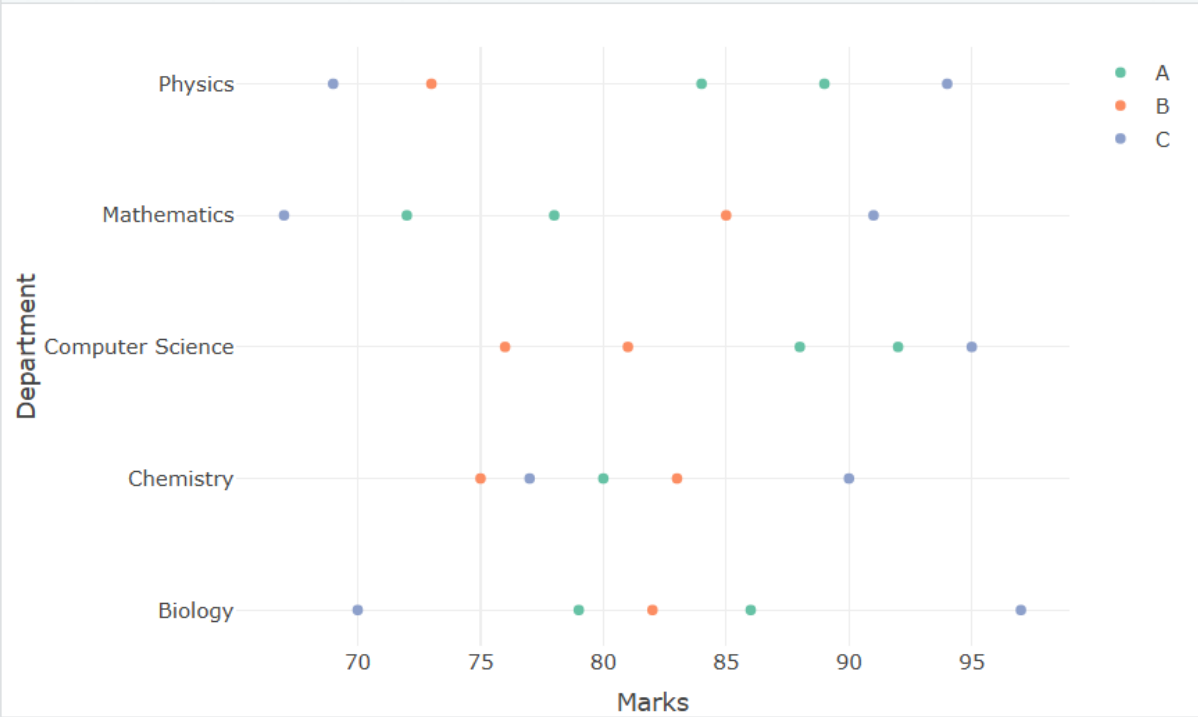
```
        mode="markers")
fig
```






Marks Spread by Department
90
80
70
Marks
Biology
Chemistry
Computer Science
Mathematics
Physics
Department
Department
Biology
Chemistry
Computer Science
Mathematics
Physics

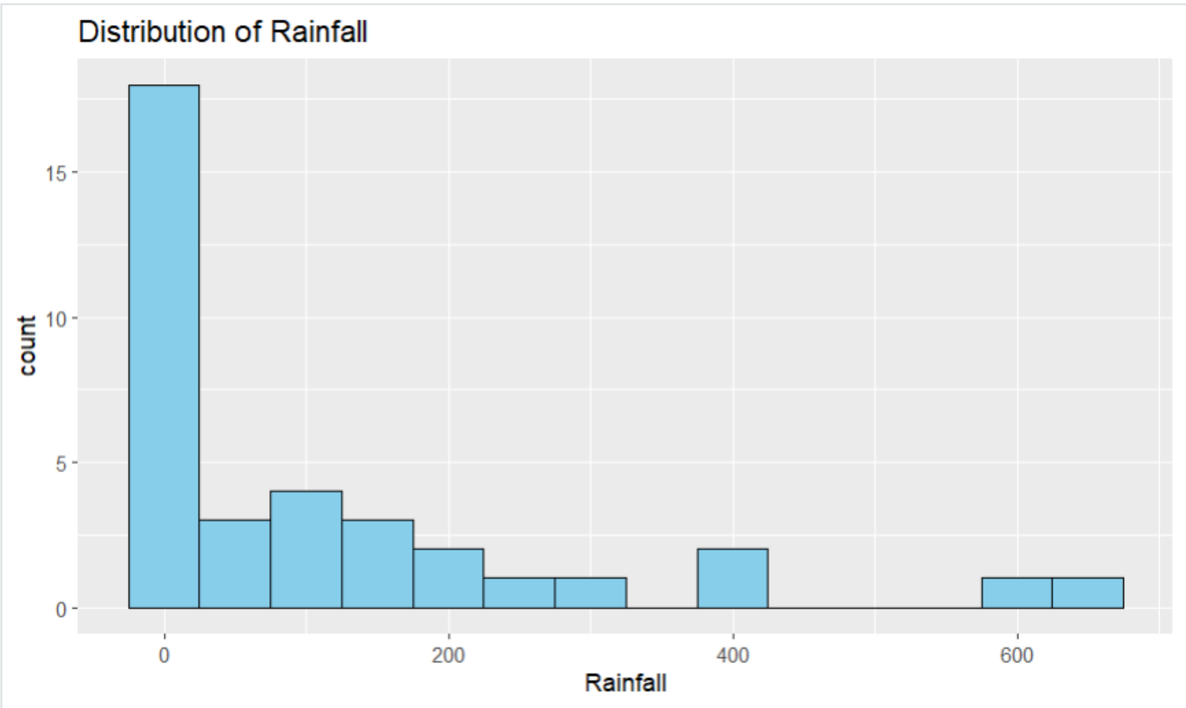

Distribution of Student Marks
5
4
3
2
1
0
count
70
80
90
Marks

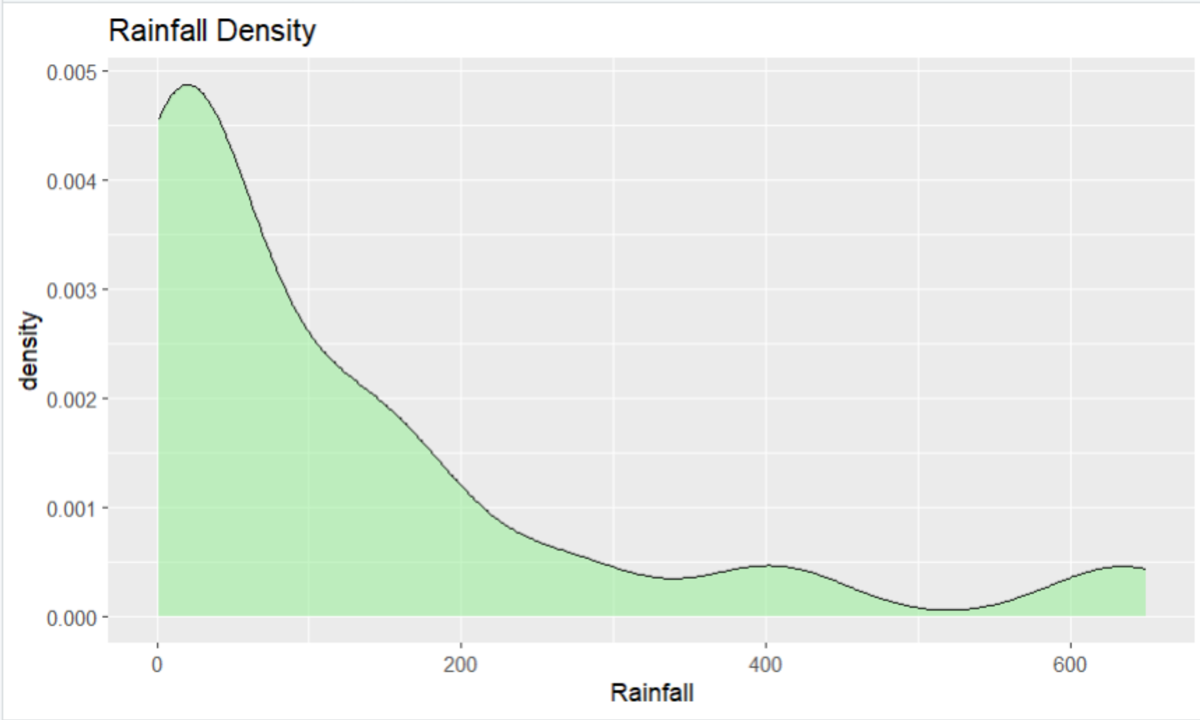


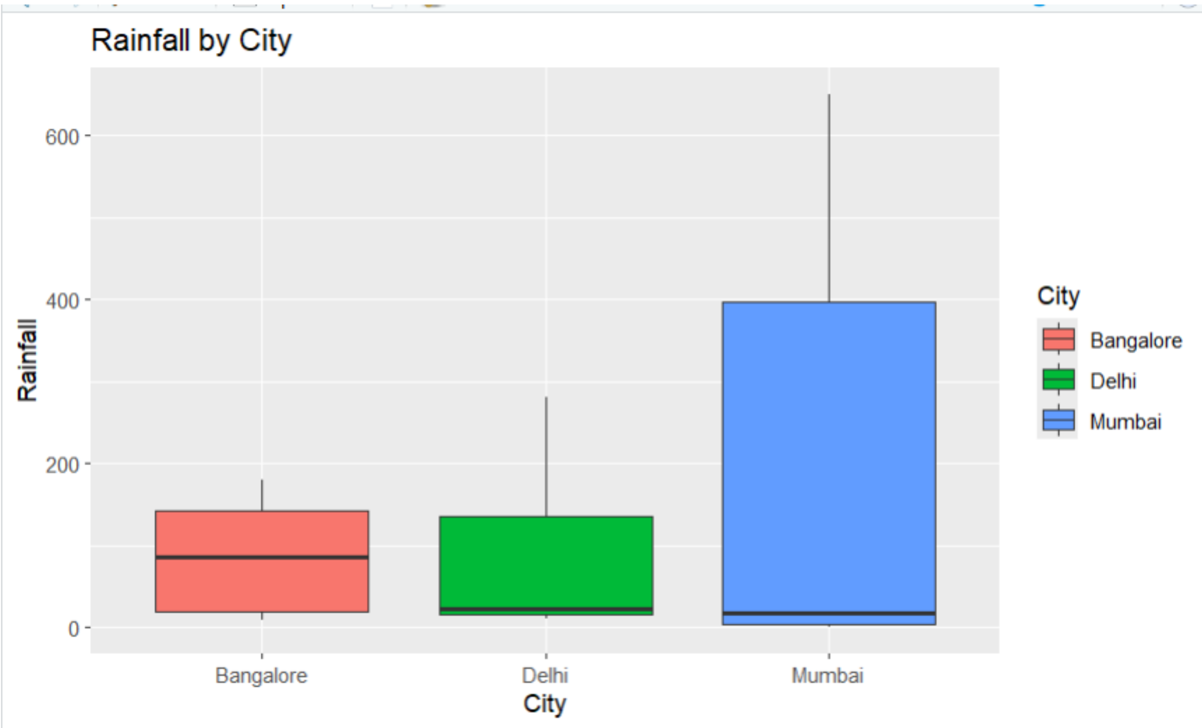
4.

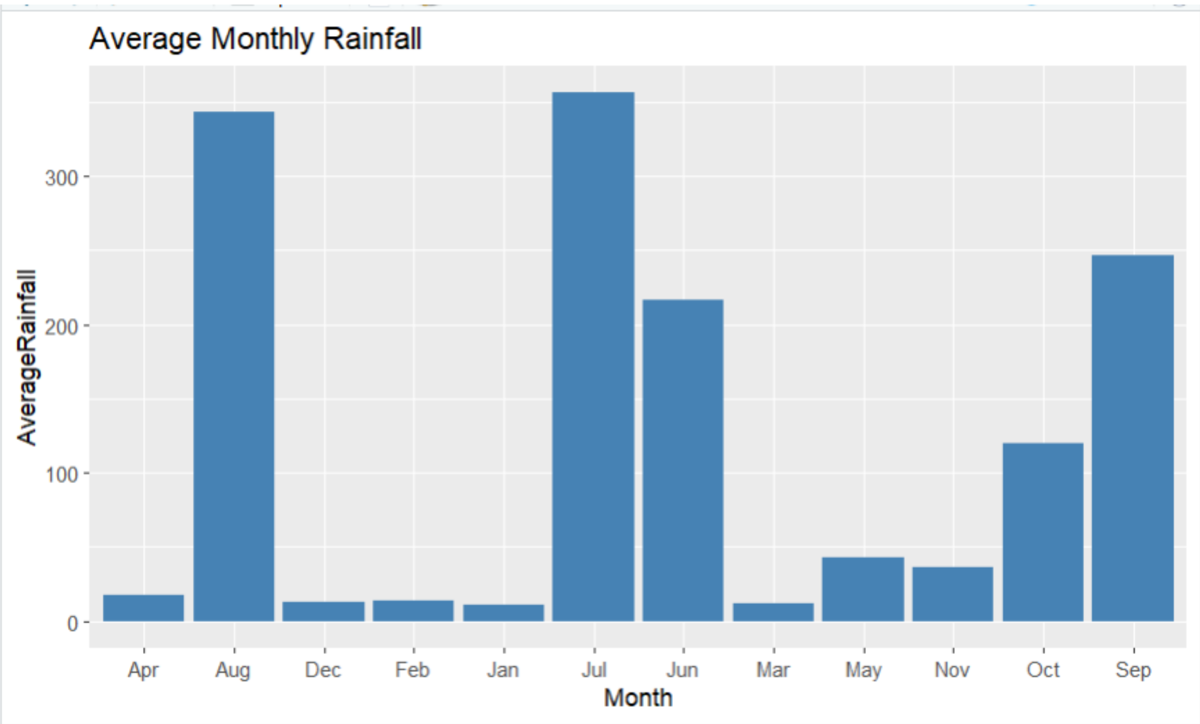
```
library(ggplot2)

library(dplyr)


# Read Dataset

df <- read.csv("climate_rainfall.csv")


# 1. Histogram

ggplot(df, aes(Rainfall)) +

 geom_histogram(binwidth=50,

         fill="skyblue",

         color="black") +

 ggtitle("Distribution of Rainfall")


# 2. Density Plot

ggplot(df, aes(Rainfall)) +

 geom_density(fill="lightgreen",

        alpha=0.5) +
```

```r
  ggtitle("Rainfall Density")

# 3. Box Plot
ggplot(df,
       aes(City, Rainfall, fill=City)) +
  geom_boxplot() +
  ggtitle("Rainfall by City")

# 4. Bar Chart
monthly <- df %>%
  group_by(Month) %>%
  summarise(AverageRainfall = mean(Rainfall))

ggplot(monthly,
       aes(Month, AverageRainfall)) +
  geom_bar(stat="identity",
           fill="steelblue") +
  ggtitle("Average Monthly Rainfall")
```




Rainfall Density
0.005
0.004
0.003
0.002
0.001
0.000
density
0
200
400
600
Rainfall

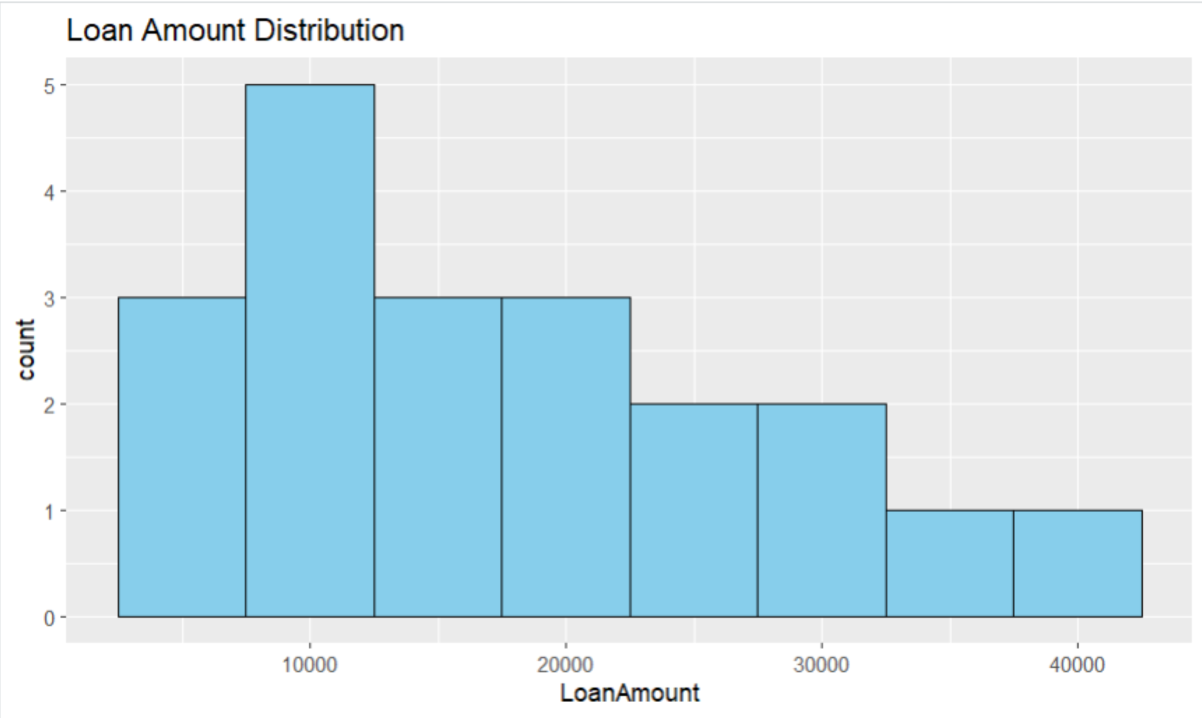

Rainfall by City
600
400
200
0
Rainfall
Bangalore
Delhi
Mumbai
City
City
Bangalore
Delhi
Mumbai

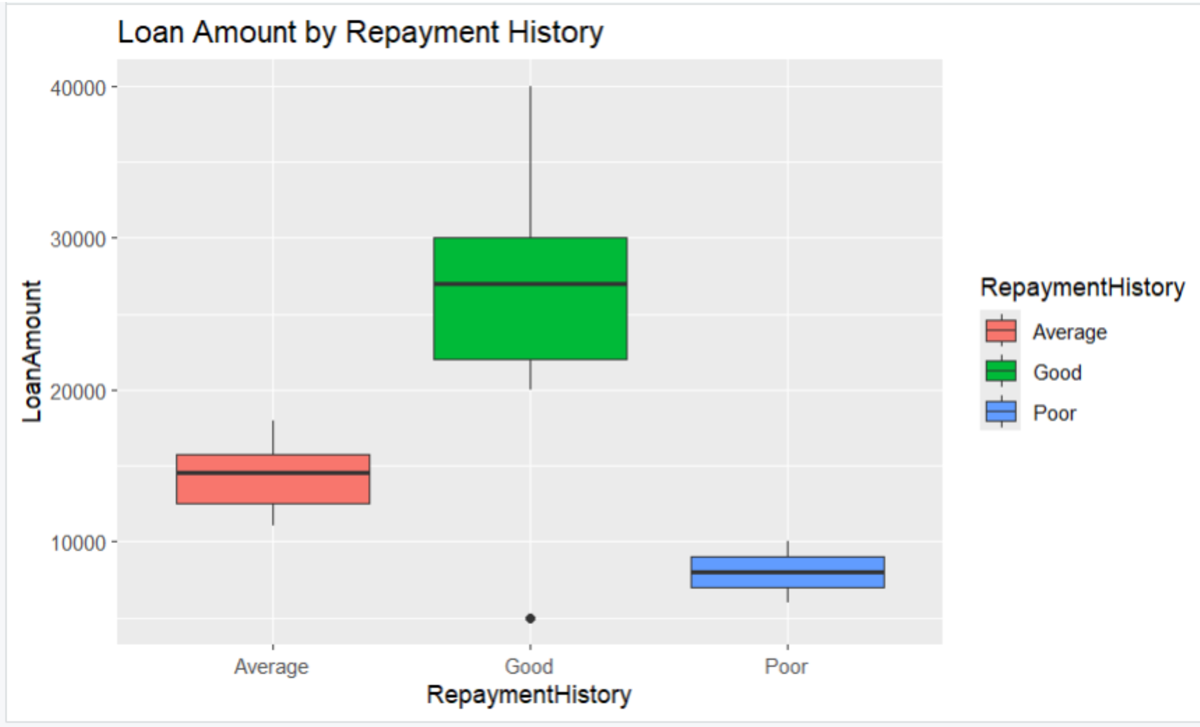


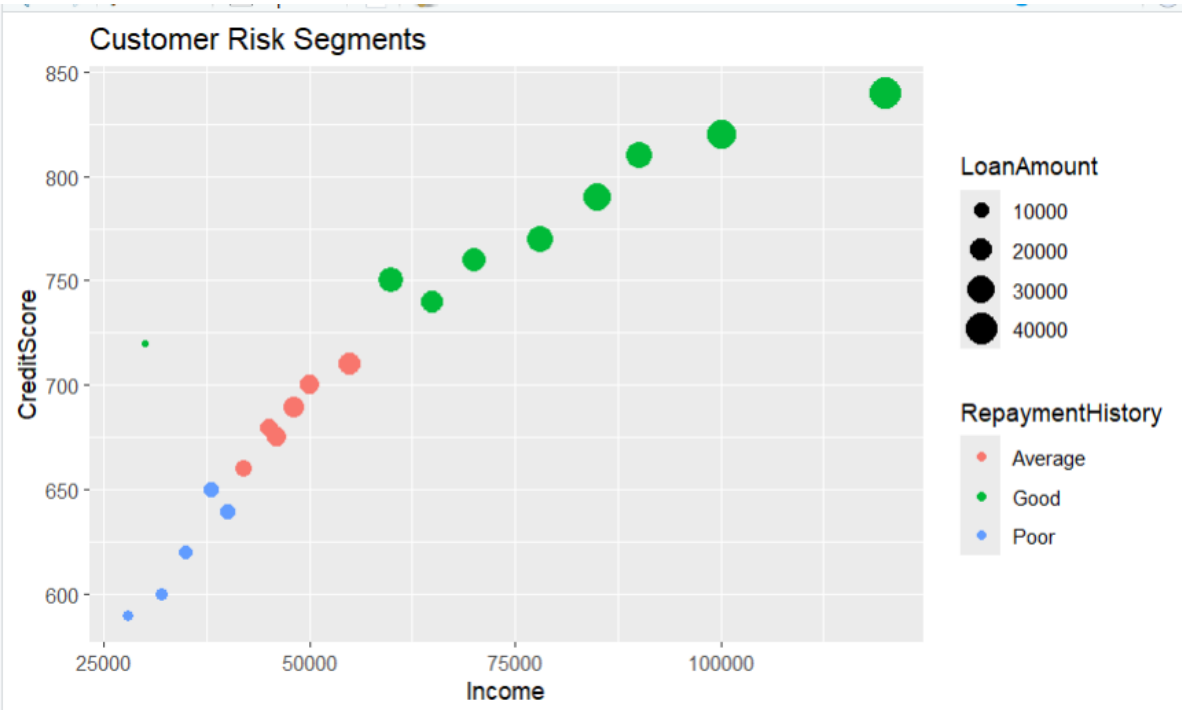
5.

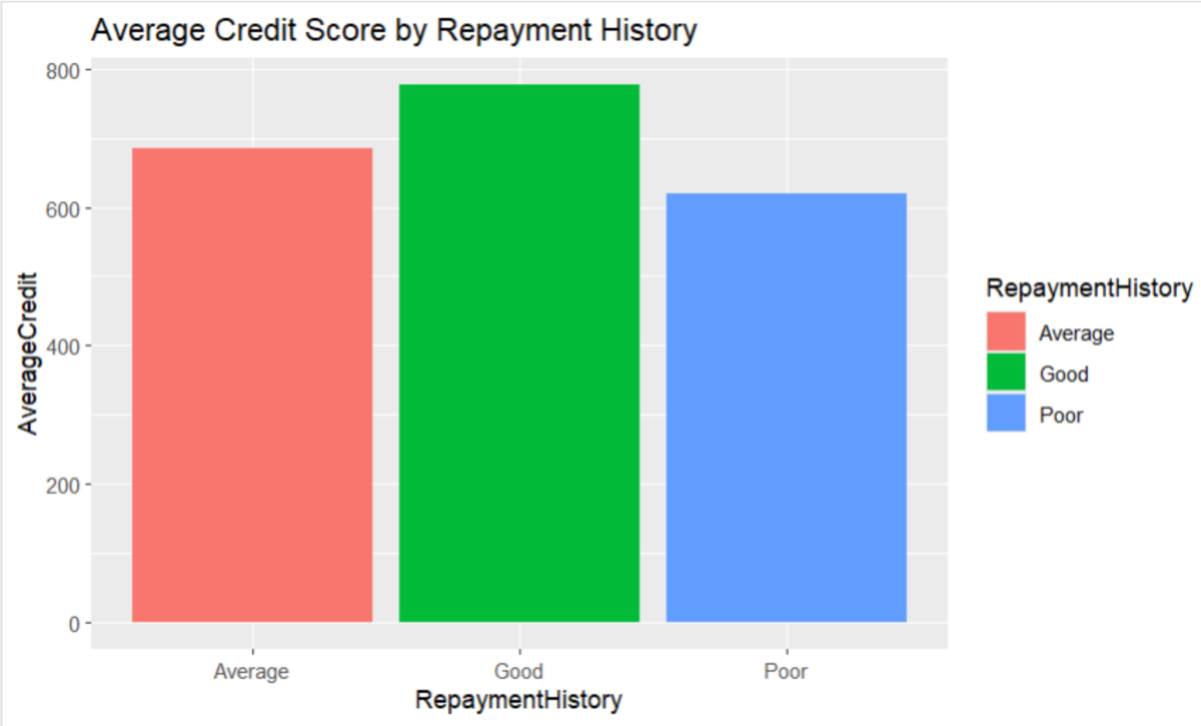
```
library(ggplot2)

library(dplyr)

# Read dataset

df <- read.csv("bank_loan_risk.csv")

# 1. Histogram

ggplot(df, aes(LoanAmount)) +

 geom_histogram(binwidth=5000,

           fill="skyblue",

           color="black") +

 ggtitle("Loan Amount Distribution")

# 2. Box Plot

ggplot(df,

     aes(RepaymentHistory,

        LoanAmount,

        fill=RepaymentHistory)) +

 geom_boxplot() +

 ggtitle("Loan Amount by Repayment History")
```

```
# 3. Scatter Plot
ggplot(df,
    aes(Income,
      CreditScore,
      color=RepaymentHistory,
      size=LoanAmount)) +
 geom_point() +
 ggtitle("Customer Risk Segments")
# 4. Bar Chart
avg_credit <- df %>%
 group_by(RepaymentHistory) %>%
 summarise(AverageCredit = mean(CreditScore))
ggplot(avg_credit,
    aes(RepaymentHistory,
      AverageCredit,
      fill=RepaymentHistory)) +
 geom_bar(stat="identity") +
 ggtitle("Average Credit Score by Repayment History")
```




Loan Amount by Repayment History
40000
30000
20000
10000
LoanAmount
Average
Good
Poor
RepaymentHistory
RepaymentHistory
Average
Good
Poor


Customer Risk Segments
850
800
750
700
650
600
CreditScore
25000
50000
75000
100000
Income
LoanAmount
10000
20000
30000
40000
RepaymentHistory
Average
Good
Poor


Average Credit Score by Repayment History
800
600
400
200
0
AverageCredit
Average
Good
Poor
RepaymentHistory
RepaymentHistory
Average
Good
Poor